

BUS 104

Decision Analysis and Management Science

Instructor:	Danko Turcic, Associate Professor of Operations Management	Quarter:	Fall 2026
Office:	Anderson-1 421	Meeting Time:	MW 11:00 AM-12:20 PM
Email:	danko.turcic@ucr.edu	Classroom:	SB 165
		Website:	https://eLearn.ucr.edu
		Office Hours (in person):	M 12:30 PM-1:30 PM (in person) Tue 11:00 AM-12:00 PM (Zoom ID: 752 601 2800); or, by appointment.

School of Business

Vision

We pursue excellence in all that we do, as our vision is to be an internationally recognized leader in business education and research.

Mission

We create knowledge through impactful research, unlock potential through exceptional educational experiences, and foster success in an innovative and collaborative environment.

Strategy

We pursue excellence in all we do by cultivating our environment, engaging our community and empowering our stakeholders.

Course Description

The course fosters analytical and quantitative approaches to solving business problems. Regardless of their functional orientation, many managerial decisions are increasingly based on analysis using quantitative models from management science. Management science tools, techniques and concepts (e.g., data, models, and software programs) have dramatically changed how businesses operate in manufacturing, services, marketing, finance, and other areas.

The primary goal of the course is to help students become effective problem solvers and intelligent business decision-makers in various management situations. The course utilizes structured problem-solving approaches heavily relying on data for defining the problem, uncovering useful relationships between critical variables and outcomes, defining measures for evaluating alternatives, and modeling underlying conceptual relationships with constraining resources. Applications of those analytic tools will be illustrated using examples from various business functional areas: finance, marketing, operations, economics and strategy, etc.

Course Objective

- Developing computer skills to solve decision analysis and management science problems
- Decisions under certainty (deterministic math programming)
- Decisions under uncertainty (simulation and decision analysis)

Course Materials

Lecture Notes	Electronic version of class notes will be periodically posted on eLearn
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Helpful References

1. B. Render, R.M. Stair Jr., and N. Balakrishnan, *Managerial Decision Modeling with Spreadsheets*, any edition, Prentice Hall
2. Hillier, F. S., and G. J. Lieberman: *Introduction to Operations Research*, McGraw-Hill Publishing Company, any edition
3. Taha, H. A.: *Operations Research: An Introduction*, Prentice Hall, any edition

Grading Policy

Component	Weight
Mid-term exam	25%
Final exam	25%
Homework	40%
Online Case	10%

Grades Scored Between		Letter Grade
97%	100%	A+
94%	97%	A
90%	94%	A-
87%	90%	B+
84%	87%	B
80%	84%	B-
77%	80%	C+
74%	77%	C
70%	74%	C-
67%	70%	D+
64%	67%	D
60%	64%	D-
0%	60%	F

Grades will be posted throughout the term on eLearn. If you find any problem with your score, you must inform the instructor within one week from the time this score is posted. After one week, scores will not be reviewed.

Case Study

Each student must purchase a Harvard Business Publishing Course Pack. (The link to this course pack will be emailed to students and posted in the course repo.) We'll be running online case study (online game) through the course pack, and only those who purchased it can receive credit for their work.

Grading for Simulation-Based Cases

For the case study, the primary objective is to maximize a performance metric (e.g., portfolio return). As such, grades are assigned based on **relative class performance** using percentile ranks. This approach rewards deep understanding of the underlying operations concepts and encourages strategic thinking in a competitive setting.

Calculation of Grades

1. All submitted simulation results are collected and sorted in descending order by the target performance metric (e.g., portfolio return).
2. Each student's **percentile rank** is computed within the class (using inclusive percentile ranking, which fairly handles ties by assigning the same or interpolated rank to students with identical results).
3. Case scores (as percentages out of 100) are then assigned according to the following percentile thresholds:

Important notes:

Percentile Rank	Case Score (%)	Approximate Class Position
95–100	100	Top ~5%
90–94.9	95	Next ~5–10%
85–89.9	91	
80–84.9	88	
75–79.9	86	Middle to upper-middle
>0–74.9	80	
<i>No scores below 80% are assigned for submitted work.</i>		

Table 1: Scoring scale for simulation-based cases

- Students who submit a valid attempt (i.e., complete the required simulation runs and submit the requested deliverables) **will receive at least 80%**, regardless of their relative rank. This reflects that meaningful engagement with the material is rewarded.
- An **F (0%)** is assigned **only** in cases of non-submission, no attempt, or failure to follow submission instructions (e.g., missing files, wrong format, or no results provided).
- Ties are handled naturally: students with identical performance receive the same percentile rank and thus the same case score. No arbitrary tie-breakers are applied to the primary metric.

Homework

Homework provides an opportunity to practice the modeling and analysis skills introduced in the course. The homework will emphasize the quantitative aspects of the course material and provide feedback on how well you are mastering the analytical techniques. The best way to learn Operations Management is to work on problems. Homework will be posted in Canvas. Each posted homework will clearly show when it is due. Homework submissions will also be handled through Canvas. Each homework assignment allows two submission attempts per student, and only the final submitted attempt will be used for grading. Late submissions get 5% per day penalty.

Exam

There will be an in-person final exam. The final examination will be held sometime between Saturday, December 5, 2026, and Friday, December 11, 2026. Individual section schedules are officially generated and solidified inside R'Web by the 7th week of the quarter. I will share the exam date with the class as soon as it is known.

Academic Integrity

At the University of California, Riverside honesty and integrity are fundamental values that guide and inform us as individuals and as a community. The academic culture requires that each student take responsibility for learning and for producing work that reflect their intellectual potential, curiosity, and capability. Students must represent themselves truthfully, claim only the work that is their own, acknowledge their use of the words, research results, and ideas of others, using the methods accepted by the appropriate academic disciplines, and participate honestly in all academic assignments. Misunderstanding of the appropriate academic conduct will not be accepted as an excuse for academic misconduct. If a student is in doubt about appropriate academic conduct in a

particular situation, he or she should consult with the instructor in the course to avoid the serious charge of academic misconduct. To ensure the highest standard of academic integrity, all students should be familiar with the guidelines posted at: <https://conduct.ucr.edu/policies/academic-integrity-policies-and-procedures>

School of Business Guidelines Regarding Generative AI:

AI is not your work. Any work produced by AI that is passed off as your own is an act of academic dishonesty and violates UCR's academic honor code.

AI usage is subject to the discretion of each instructor. Each instructor will have specific guidelines on AI usage that must be adhered to for their specific class.

AI is not perfect. Its responses may contain inaccuracies, and it cannot critically evaluate the content it produces; you need to critically evaluate any output.

BUS 104 AI CLASS POLICY

The use of generative AI tools (e.g., ChatGPT, Gemini) is permitted for the following:

- Brainstorming and refining your ideas.
- Writing code
- Drafting an outline to organize your thoughts.

The use of generative AI tools is not permitted for the following:

- Generating entire answers to homework assignments (writing code to solve assignment problems using AI is okay)
- Generating answers on exams.

Course Schedule¹

MONDAY		WEDNESDAY	
Sep 28th	1	30th	2
Course Introduction • Introductions • Suggested Reading: §1 in the textbook, and "Operation Everything"		Introduction to Spreadsheet-Based Optimization • Case Study: "The Diet Problem" • Suggested Reading: Appendix – Using Excel Commands (We will be using Google Sheets in this class, but Excel is an excellent alternative.)	

¹This is an expected course schedule. I reserve the right to make minor changes and updates throughout the quarter.

MONDAY		WEDNESDAY	
Oct 5th	3	7th	4
Introduction to Python • Data structures and basic commands • Using Gemini to generate code		Introduction to Python-Based Optimization • Revisiting “The Diet Problem”	
12th	5	14th	6
Formulating Linear Optimization Problems • “Product Mix,” “Blending,” and “Staffing” problems • Suggested Reading: §2 in the textbook		Linear Optimization (Continued) • “Transshipment” and “Assignment” problems	
19th	7	21st	8
Solving Linear Optimization Google Sheets • Solving “Product Mix” and “Staffing” problems using Sheets		Solving Linear Optimization Python • Solving “Product Mix” and “Staffing” problems using Python	
26th	9	28th	10
Robustness of Math Programming Solutions • Introduction to sensitivity analysis: Shadow prices and reduced costs • “Parker Sisters” case study		Integer Programming Introduction • Finish “Parker Sisters” • “The Knapsack” problem • Suggested Reading: §5 in the textbook	
Nov 2nd	11	4th Midterm Exam (exam will be administered online through Canvas)	
Integer Programming Problem Formulations and Solutions • “Logical constraints in 0-1 Problems” and “Production Thresholds”			

MONDAY		WEDNESDAY	
9th	12	11th	14
Solving Integer Programs • “Logical constraints in 0-1 Problems” and “Production Thresholds” • Google Sheets and Python		Veterans Day	
16th	13	18th	14
Introduction to Nonlinear Optimization • Formulating nonlinear problems • Suggested Reading: §5 in the textbook		Nonlinear Optimization • Formulating nonlinear programs from data • Solving nonlinear programs in Python	
23rd	15	25th	16
Introduction to Simulation Google Sheets and Python • Generating random numbers • Suggested Reading: §9, and Appendix (Probability Concepts and Applications) in the textbook		Simulation Exercises Google Sheets and Python • Simulating queues in Google Sheets • Newsvendor problem in Python	
30th	17	Dec 2nd	18
Simulation Exercises Confidence Intervals • Analyzing simulation output		Wrap-Up • Finishing up any unsolved problems from previous classes • Final exam review	